Investigations and the role of regression modelling

Data scientists don't do statistics just for fun (although, clearly statistics is indeed fun!). At the heart of each data science project is a question to be answered.

This section of the notes begins by thinking about the different types of investigation that might be carried out within a data science project. We consider three important classes of investigation type: **description**, **prediction** and **causal**.

We then move on to consider a commonly used family of statistical analysis: **regression modelling**. Three sessions introduce linear regression, beginning with the simplest type which we call **simple linear regression** involving a single explanatory variable. We then extend this to incorporate multiple explanatory variables, through **multivariable linear regression** modelling. We explore how to model various types of explanatory variables, including continuous, binary and categorical covariates and discover how to include interactions and higher-order terms (which are need to model non-linear relationships) in the regression model. The last of the linear regression sessions explores diagnostics to assess whether the underlying assumptions of the linear model hold in a particular dataset.

These ideas are then extended to other settings in the remaining two sessions. First, we meet **logistic regression**, an extension of linear regression modelling to settings where the outcome variable is binary. Finally, we define the **Generalised Linear Model (GLM)**, which is a generalisation of linear regression to a wide range of settings and can be seen as a way of unifying linear, logistic and Poisson regression models, as well as many other types of regression model. We explore **Poisson regression** as an important example of a GLM.

We conclude this section of the notes by returning to the idea of investigations and – armed with our new knowledge about regression modelling – consider the role of regression modelling in different types of investigations.